

Ex. No: 10 Paging Technique

C PROGRAM

```
#include <stdio.h>

int main()
{
    int pageTable[20];
    int pageSize;
    int numPages;
    int logicalAddress;
    int pageNumber;
    int offset;
    int frameNumber;
    int physicalAddress;
    int i;

    printf("Enter Page Size: ");
    scanf("%d", &pageSize);
    printf("Enter Number of Pages: ");
    scanf("%d", &numPages);
    printf("Enter Frame Numbers for Each Page:\n");
    for(i = 0; i < numPages; i++)
    {
        printf("Page %d -> Frame: ", i);
        scanf("%d", &pageTable[i]);
    }
    printf("Enter Logical Address: ");
```

```

scanf("%d", &logicalAddress);

pageNumber = logicalAddress / pageSize;

offset = logicalAddress % pageSize;

if(pageNumber >= numPages)
{
printf("Invalid Logical Address\n");

return 0;
}

frameNumber = pageTable[pageNumber];

physicalAddress = (frameNumber * pageSize) + offset;

printf("\nPage Number : %d", pageNumber);

printf("\nOffset : %d", offset);

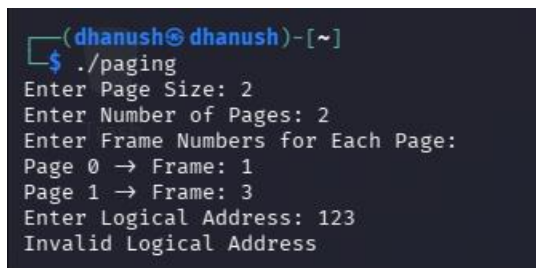
printf("\nFrame Number : %d", frameNumber);

printf("\nPhysical Address : %d\n", physicalAddress);

return 0;
}

```

Output:



```

(dhanush@dhanush)-[~]
$ ./paging
Enter Page Size: 2
Enter Number of Pages: 2
Enter Frame Numbers for Each Page:
Page 0 → Frame: 1
Page 1 → Frame: 3
Enter Logical Address: 123
Invalid Logical Address

```

SHELL SCRIPT

```
#!/bin/bash
```

```
echo "Enter Page Size:"
read pageSize
echo "Enter Number of Pages:"
read numPages
declare -a pageTable
for ((i=0;i<numPages;i++))
do
echo "Enter Frame Number for Page $i:"
read pageTable[$i]
done
echo "Enter Logical Address:"
read logicalAddress
pageNumber=$((logicalAddress / pageSize))
offset=$((logicalAddress % pageSize))
if [ $pageNumber -ge $numPages ]
then
echo "Invalid Logical Address"
exit
fi
frameNumber=${pageTable[$pageNumber]}
physicalAddress=$((frameNumber * pageSize + offset))
echo "Page Number : $pageNumber"
echo "Offset : $offset"
echo "Frame Number : $frameNumber"
echo "Physical Address : $physicalAddress"
```

Output:

```
(dhanush@dhanush)-[~]  
$ nano paging.sh  
  
(dhanush@dhanush)-[~]  
$ chmod +x paging.sh  
  
(dhanush@dhanush)-[~]  
$ bash paging.sh  
Enter Page Size:  
2  
Enter Number of Pages:  
2  
Enter Frame Number for Page 0:  
1  
Enter Frame Number for Page 1:  
23  
Enter Logical Address:  
12344  
Invalid Logical Address
```